CODE BASICS

SQL THROUGH ONE DATASET

DAY 13 / 15

08/08/26

Forecast accuracy came out suspiciously high

Two LEFT JOINS fixed every missing month

ATLIQ HARDWARE GDB0041

MYSQL | Vishnu Ram Sachin | Data Analyst

Problem
Statement

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- Yesterday : Fixed running total that broke at the fiscal year boundary

- Today Leadership wants an honest forecast vs actuals comparison per product per month

The Trap ❌

- INNER JOIN on date , customer_code and product_code looks clean but it only keeps month present on both sides

Forecast , Zero sold , Dropped

Sold , Never forecast , Dropped

Requirment

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Keeps every forecasted month , even with Zero sales

Keep every sold month , even with Zero forecast

Two LEFT JOINs , one filter not a single INNER JOIN

The Fix

```
FROM fact_sales_monthly fsm
LEFT JOIN fact_forecast_monthly ffm
        UNION ALL
FROM fact_forecast_monthly ffm
LEFT JOIN fact_sales_monthly fsm
WHERE fsm.product_code IS NULL
```

A Quick Analogy

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A Party's Guest list vs It's Sign in sheet

Invited but didn't come

We expected them and sent the invite. They never showed up

(forecast LEFT JOIN, WHERE sale IS NULL)

Not invited , Came anyway

We never invited them , but they showed up and signed in

(sales LEFT JOIN , forecast)

The Full Picture

Both grouped combined with everyone who matched ,
Nobody counted twice

(UNION ALL)

Fix Query

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```
WITH forecast_actual_compare_report AS (
   SELECT
  fsm.date,
                  MONTHNAME(fsm.date)  AS  month_name  ,  fsm.customer_code,
fsm.product_code,
      fsm.fiscal_year,
  COALESCE(ffm.forecast_quantity, 0) AS forecast_quantity,
  fsm.sold_quantity
   FROM fact_sales_monthly AS fsm
  LEFT JOIN fact_forecast_monthly AS ffm
   ON fsm.date = ffm.date
   AND fsm.customer_code = ffm.customer_code
   AND fsm.product_code = ffm.product_code
```

Fix Query

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```
UNION ALL

SELECT
    ffm.date,
    MONTHNAME(ffm.date) AS month_name, ffm.customer_code,
    ffm.fiscal_year,
    ffm.product_code, ffm.forecast_quantity, 0 AS sold_quantity
FROM fact_forecast_monthly AS ffm
LEFT JOIN fact_sales_monthly AS fsm
 ON ffm.date = fsm.date
 AND ffm.customer_code = fsm.customer_code
 AND ffm.product_code = fsm.product_code
WHERE fsm.product_code IS NULL
)
SELECT
facr.month_name, facr.fiscal_year,
facr.customer_code, facr.product_code,
facr.forecast_quantity, facr.sold_quantity,
(forecast_quantity - sold_quantity) AS forecast_error
FROM forecast_actual_compare_report facr ;
```

Results

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```sql
WITH forecast_actual_compare_report AS (
    SELECT
        fsm.date,
        MONTHNAME(fsm.date) AS month_name , fsm.customer_code, fsm.product_code,
        fsm.fiscal_year,
        COALESCE(ffm.forecast_quantity, 0) AS forecast_quantity,
        fsm.sold_quantity
    FROM fact_sales_monthly AS fsm
        LEFT JOIN fact_forecast_monthly AS ffm
            ON fsm.date = ffm.date
            AND fsm.customer_code = ffm.customer_code
            AND fsm.product_code = ffm.product_code

    UNION ALL

    SELECT
        ffm.date,
        MONTHNAME(ffm.date) AS month_name, ffm.customer_code,
```

month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
September	2018	70002017	A0118150101	18	51	-33
September	2018	70002018	A0118150101	11	77	-66
September	2018	70003181	A0118150101	9	17	-8
September	2018	70003182	A0118150101	6	6	0

Results

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fact_sales_monthly | dim_customer | fact_gross_price | fact_forecast_monthly | SQL File 11*

Limit to 50000 rows

```sql
    UNION ALL

    SELECT
        ffm.date,
        MONTHNAME(ffm.date) AS month_name, ffm.customer_code,
        ffm.fiscal_year,
        ffm.product_code, ffm.forecast_quantity, 0 AS sold_quantity
    FROM fact_forecast_monthly AS ffm
        LEFT JOIN fact_sales_monthly AS fsm
            ON ffm.date = fsm.date
            AND ffm.customer_code = fsm.customer_code
            AND ffm.product_code = fsm.product_code
    WHERE fsm.product_code IS NULL
)
SELECT
facr.month_name, facr.fiscal_year,
facr.customer_code, facr.product_code,
facr.forecast_quantity, facr.sold_quantity,
(forecast_quantity - sold_quantity) AS forecast_error
FROM forecast_actual_compare_report facr ;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
September	2018	70002017	A0118150101	18	51	-33
September	2018	70002018	A0118150101	11	77	-66
September	2018	70003181	A0118150101	9	17	-8

Result 1 | Read Only

Results

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fact_sales_monthly | dim_customer | fact_gross_price | fact_forecast_monthly | SQL File 11*

Limit to 50000 rows

```
    UNION ALL

    SELECT
        ffm.date,
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

month_name	fiscal_year	customer_code	product_code	forecast_quantity	sold_quantity	forecast_error
September	2018	70002017	A0118150101	18	51	-33
September	2018	70002018	A0118150101	11	77	-66
September	2018	70003181	A0118150101	9	17	-8
September	2018	70003182	A0118150101	6	6	0
September	2018	70006157	A0118150101	5	5	0
September	2018	70006158	A0118150101	6	7	-1
September	2018	70007198	A0118150101	4	29	-25
September	2018	70007199	A0118150101	7	34	-27
September	2018	70008169	A0118150101	7	22	-15
September	2018	70008170	A0118150101	8	5	3
September	2018	70011193	A0118150101	5	10	-5
September	2018	70011194	A0118150101	7	4	3
September	2018	70012042	A0118150101	0	0	0
September	2018	70012043	A0118150101	0	0	0
September	2018	70013125	A0118150101	2	1	1
September	2018	70013126	A0118150101	2	1	1
September	2018	70016178	A0118150101	0	1	-1
September	2018	70022085	A0118150101	12	20	-8

Result 1 | Read Only

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| Vishnu Ram Sachin | Data Analyst

SQL THROUGH ONE DATASET CHALLENGE